Stock Price Prediction Report

S&P 500 Index Analysis

Date: April 20, 2026

Project Source Code (Google Colab):
https://colab.research.google.com/drive/1BtWOxBdtKYj1wcn9QQH1ouulu3ro5RJc?usp=sharing

1. Introduction

The objective of this project is to develop a predictive model capable of forecasting the daily price movements of the S&P 500 index. By treating the market as a classification problem—predicting whether the price will increase or decrease—this study utilizes historical data to identify recurring patterns.

2. Description of Dataset

The dataset consists of historical daily trading records for the S&P 500 index. Key features utilized include Open, High, Low, Close prices, and Volume. These are processed to create a binary target where '1' indicates an upward price movement.

3. Methodology

The analytical workflow followed a structured machine learning pipeline. In the development phase, three different algorithms were evaluated to determine which provided the most reliable signals. After comprehensive testing and performance evaluation, the best-performing model was selected for the final forecasting tool (Random Forest). A walk-forward backtesting approach was employed, training on an expanding window of data and testing in 250-day increments to ensure realistic, bias-free results.

4. Results & Performance Analysis

The final model was evaluated based on the outcome of 'Price Up' predictions. The analysis reveals that the model achieved over 500 correct predictions (Price went Up) against 376 incorrect predictions (Price went Down).

Metric	Observation
Model Precision	The model correctly predicted price increases more frequently than it failed, establishing a profitable signal-to-noise ratio.
Prediction Accuracy	While demonstrating strong predictive power during stable phases, the model faced challenges during recent high-volatility shifts in April 2026.

5. Interpretation & Conclusion

By testing multiple algorithms and selecting the optimal performer, this project successfully captures broad market trends. The results demonstrate that correct 'Buy' signals significantly outweigh incorrect ones. However, the missed signals in mid-April suggest that future iterations should incorporate external macroeconomic indicators to further enhance precision.

6. Model Deployment

The finalized Random Forest model has been successfully deployed as an interactive web application. This allows users to generate real-time forecasts and visualize historical S&P 500 performance directly through a browser.

- **Deployment Platform:** Streamlit Cloud.
- **Live Application Link:** https://da380project-vdddwvejysotpzxmcwd9uw.streamlit.app/
- **Core Functionality:** The app fetches daily trading data (Open, High, Low, Close, and Volume) and processes it through the machine learning pipeline to provide binary "Price Up" signals.

7. Final Summary

This project transitioned from a historical data analysis in Google Colab to a functional trading tool. While the model currently achieves a positive signal-to-noise ratio—with over 500 correct upward predictions—the live deployment serves as a foundation for future updates, such as the inclusion of macroeconomic indicators to better handle high-volatility shifts.